

Multiobjective optimization of carbon fiber–reinforced epoxy composite properties using nonparametric modeling

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ABSTRACT

Carbon fiber–reinforced epoxy composites are widely used in the marine and aviation sectors due to their superior mechanical properties. As research into these materials progresses, the demand for materials with even more complex and superior properties has increased. Traditional trial-and-error methods are limited to optimizing single factors and are plagued by heavy workloads and prohibitive costs when applied to multifactor composite designs. Addressing these problems, this study introduces a nonparametric multiobjective optimization process for composite materials. This approach employs a Gaussian process (GP) to establish a multiobjective regression model and uses nondominated sorting genetic algorithm II (NSGA II) to determine the Pareto front. Pareto solutions were ranked in accordance with their similarity to ideal solutions and on the basis of predetermined preference weights. The results demonstrate variations in optimal outcomes contingent on the weight values assigned. The differences between the optimized results and experimental validations reached 12.9%, with the minimum deviation being 0.6%. The effectiveness of the proposed method was demonstrated. The strategy combines GP and NSGA II to extend design methods for small samples and materials with multiple attributes. This method enables the simultaneous optimization of multiple conflicting objectives, offering greater efficiency than addressing them individually.

1. Introduction

Epoxy resin (EP) is widely used in automotive, marine, aerospace, and biomedical industries due to its adhesive characteristics, stability, chemical resistance, and high mechanical strength [1–5]. The brittleness, limited impact resistance, susceptibility to cracking, and reduced toughness of pure EP limit its application scope. Several methods have been proposed to improve the performance of EP, including chemical copolymerization [6], utilization of hyperbranched polymers [7], and the addition of fillers [8–12]. In addition, efforts to toughen EP involve introducing a core-shell structure [13], an interpenetrating network [14], and a liquid crystal polymer structure [15,16]. Incorporating carbon fibers into resin matrix composites is an effective means of enhancing the performance of the composite. Carbon fibers are renowned for being lightweight, possessing high specific strength and modulus, and demonstrating fatigue resistance [17]. The integration of carbon fibers and EP has led to the widespread use of carbon fiber–reinforced epoxy composites in various fields. Carbon fiber–reinforced epoxy composites are strong, have high specific stiffness, and are lightweight

[18–20]. The process by which carbon fiber–reinforced epoxy systems are cured is complex and variable. Different curing process parameters can lead to differences in the properties of carbon fiber–reinforced epoxy systems. Therefore, designing and optimizing the parameters of carbon fiber–reinforced epoxy systems for different application scenarios is essential.

Optimization of carbon fiber–reinforced epoxy composites is a popular topic of research. Capela et al. [21] studied the failure mechanisms of short fiber composites with high fiber volume fractions, achieving composites with high tensile strength and stiffness through experimentation. Okabe et al. [22] investigated the correlation between differences in epoxy resin curing systems and their curing characteristics, using experimental and molecular simulation approaches. Yu et al. [23] synthesized a phosphorus and silica synergistic multifunctional epoxy additive that significantly improved the mechanical properties, flame retardancy, and water resistance of epoxy resins. The aforementioned studies relied on trial-and-error research methods. Traditional experimental methods, especially trial-and-error methods, require a significant number of experiments, resulting in the waste of time and samples.

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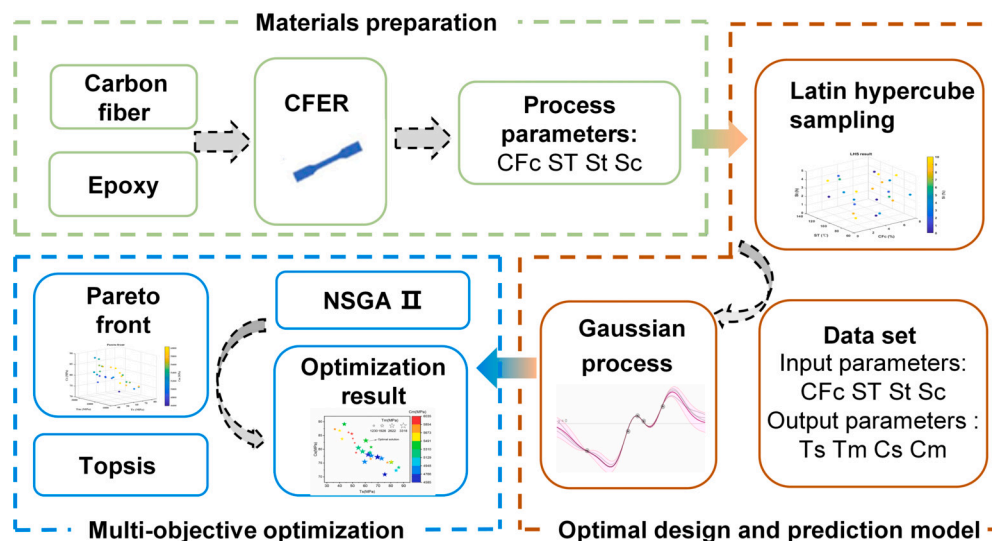


Fig. 1. Schematic of nonparametric modeling and multiobjective optimization.

Machine learning methods have increasingly been applied to material design and optimization [24–26] and can significantly reduce the experimental workload required for the design and development of new materials. Qiu et al. [27] used machine learning and deep learning to construct a model that accurately predicts the viscosity of epoxy resin. Taniguchi et al. [28] developed a system for manufacturing epoxy resin by using machine learning and determining the composition of material properties. Jin et al. [29] developed a resin system design methodology by combining molecular dynamics simulations and machine learning. The method enhanced the ultimate tensile strength of the designed resin by 32.7% and increased specific elongation by 85.5%. Nevertheless, as a data processing method, machine learning still relies on foundational data to be effective. In addition, most research focuses on optimizing either a single objective or multiple parameters. At present, multi-objective optimization is more widely used to optimize the benefits and costs of composite structures [30] and to assist multi-objective optimization and multi-criteria decision frameworks through machine learning [31]. Multiobjective optimizations that simultaneously address multiple properties are lacking in the literature.

This paper proposes a multiobjective optimization method based on a Gaussian process (GP). GPs are nonparametric function estimation models that are well-suited for solving complex problems involving high-dimensional, nonlinear, small-sample data. In this paper, a small-sample data set obtained from experiments was used as the foundational data set. A GP was used to establish the response relationship between parameters and the performance targeted for optimization. Employing the proposed method, multiobjective optimization was conducted using nondominated sorting genetic algorithm II (NSGA II). For decision-making, the technique for order preference by similarity to ideal solution (TOPSIS) method was applied.

2. Materials and methods

This paper combines machine learning with material design to predict and optimize the properties of carbon fiber–reinforced composites using small-sample data. Fig. 1 illustrates the workflow and experimental design of the proposed method. Section 2.1 details the materials involved, the composite preparation process, and testing of mechanical properties. Section 2.2 elaborates on the Latin hypercube design and Gaussian prediction model. Section 2.3 introduces the multiobjective optimization method. NSGA II was used to obtain the Pareto optimal solution set. The TOPSIS method was then used to condense multidimensional data into single-dimensional data for comparison to determine the optimal solution.

2.1. Materials and preparation

2.1.1. Materials

E-51 (epoxy equivalent 184–195 g/mol) was obtained from Nantong Xingchen Synthetic Material (Jiangsu, China). The curing agent was triethylenetetramine (TETA, CP, 68%), which was obtained from Aladdin Reagent (China). Silane coupling agent (KH-540) was obtained from China National Medicines (China). Carbon fibers were supplied by Toray Carbon Fiber (Guangdong, China). The carbon fibers had diameters ranging from 4 to 8 microns and lengths ranging from 50 to 70 microns. All materials were used as received.

2.1.2. Preparation of samples

First, a mixed solution of acetone and ethanol in a 1:1 ratio was prepared. The carbon fibers were then thoroughly immersed in this solution and treated in an ultrasonic cleaner for 2 h. Subsequently, a 5 wt% hydrogen peroxide solution was prepared using deionized water. The cleaned carbon fibers were soaked in this solution and agitated in the ultrasonic instrument for 20 min. After treatment, the surface-treated carbon fibers were repeatedly rinsed with deionized water until the wash solution reached a pH of 7. These treated carbon fibers were then mixed with the silane solution and reacted for 2 h. The silane solution comprised 90% ethanol, 5% distilled water, and 5% silane coupling agent. The solution was stirred for 1 h before use. Following the reaction, the carbon fibers were rinsed several times with deionized water to remove any residual silane coupling agent and then dried.

The reaction between carbon fibers and the silane coupling agent is illustrated in Fig. 2(b). The silane coupling agent reacted with the hydroxyl and carboxyl groups present on the carbon fiber surface to form Si–O bonds.

The EP and the triethylenetetramine were mixed in a 100:15 mass ratio. The treated carbon fibers were then added to the mixture and stirred at 500 rev/min for 5 min under a vacuum to form a resin. The resin was then injected into a mold, cured for 6 h at room temperature, and baked in a drying oven. The baking duration and temperature were determined in accordance with experimental parameters.

Carbon fiber–reinforced epoxy (CFRE) composites were prepared as depicted in Fig. 2(a). The treatment of carbon fiber surfaces involved cleaning, oxidation, and amination. The carbon fibers and epoxy were evenly mixed and poured into a mold to fabricate CFRE samples through a curing process. Specifically, epoxy resin (EP), untreated carbon fiber–reinforced epoxy composites (CFRE-P), carbon fiber–reinforced epoxy composites with silane coupling agent (CFRE-Si), and oxidized carbon

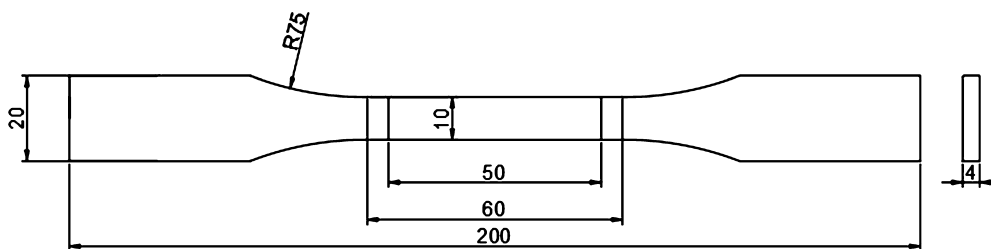
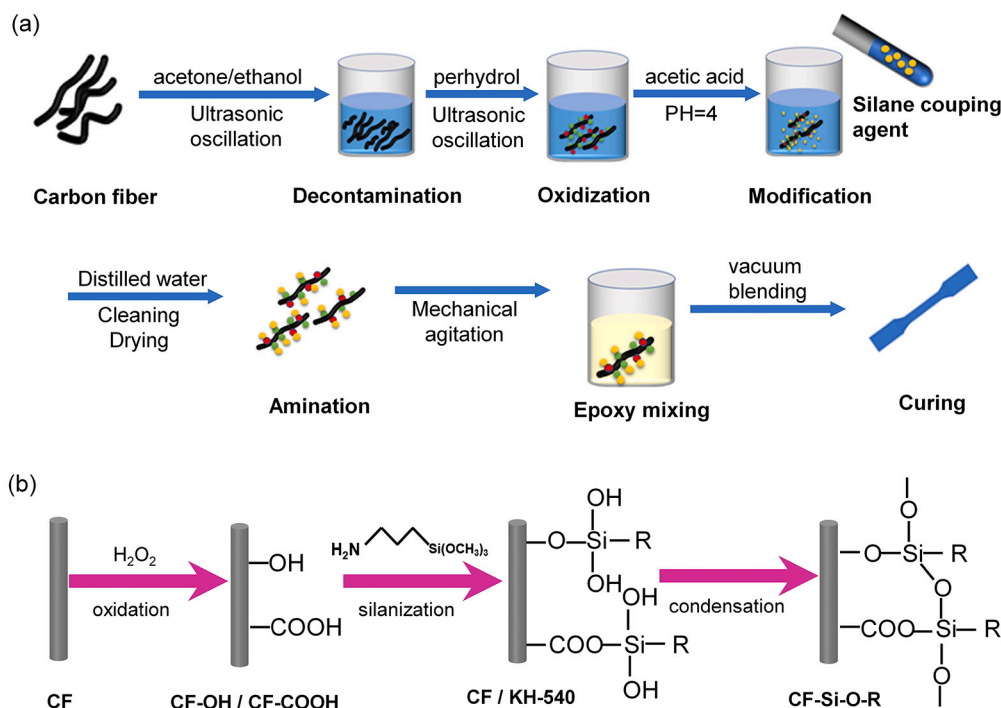


Fig. 3. Size of dumbbell-shaped specimens.

fiber-reinforced epoxy composites with silane coupling agent (CFRE-O-Si).

2.1.3. Characterization of mechanical properties

Tensile tests were conducted using a universal testing machine. In accordance with GB/T 2567-2021, these tests were performed at a quasi-static crosshead loading speed of 2 mm/min. The test was conducted at ambient temperature, utilizing dumbbell-shaped specimens measuring 200 mm × 10 mm × 4 mm in total size and an effective dimension of 50 mm × 10 mm × 4 mm. The schematic of the dumbbell specimen is shown in Fig. 3. Tests were conducted in triplicate for each test condition to ensure consistency.

Similarly, compression tests were conducted on a universal testing machine following the guidelines of GB/T 2567-2021. These tests were performed at a constant speed of 2 mm/min. The experiment was conducted at room temperature, and the test specimens were rectangular with dimensions of 35 mm × 10 mm × 10 mm. Tests were conducted in triplicate for each test condition to validate the results.

2.2. Optimal design and prediction model

2.2.1. Experimental design strategy

Latin hypercube sampling (LHS) is a method used for distributing K design points within an L-dimensional design space. This approach ensures that overlaps are avoided when projections are made onto individual dimensions [32,33]. LHS presents a considerable improvement

over the fully factorial design of experiments, which requires a comprehensive enumeration of all potential combinations across dimensions, resulting in a prohibitive number of experiments equal to the number of levels raised to the power of the number of dimensions. Additionally, LHS offers advantages over random sampling by preventing the occurrence of overlapping points.

Specifically, LHS defines an L-dimensional design space with K samples. Each dimension is divided into I bins, within which samples can be positioned. For a two-dimensional space, the distribution of points is managed so that only one point occupies any given row or column. This principle is applicable and extends to higher dimensions. Although these rules determine the placement of samples within bins, they do not specify the exact location within each bin. In practice, points can be assigned either randomly and sequentially or through a series of swaps to guarantee maximal separation among sampled points. This study used a four-sample, four-dimensional Latin hypercube design optimized for our experimental needs.

2.2.2. Gaussian process

GPs comprise a collection of random variables that are jointly Gaussian-distributed when any finite subset is considered. This characteristic implies that the values of a function at each individual point in the input space of a GP conform to a Gaussian (or normal) distribution. Analogous to how a Gaussian distribution characterizes the probability distribution of individual events using mean and covariance, a GP is entirely defined by its mean function and covariance matrix. In the

context of data regression problems, GPs are used to predict the distribution of functions by being fit to a given data set [34,35].

Consider a series of material performance variables, each denoted by y_i , that vary in response to a corresponding four-dimensional process parameter x_i , with the relationship represented as a set of ordered pairs $(x_1, y_1), \dots, (x_n, y_n)$. The relationship between process parameters and material performance can be described as $y = f(x) + \varepsilon$, where ε follows a normal distribution $N(0, \sigma_n^2)$ with a mean of 0 and a variance of σ_n^2 . The function $f(x)$ is modeled as a GP, defined by its mean function $m(x)$ and covariance function $K_f(x, x')$.

$$f(x) \sim GP(m(x), K_f(x, x')) \quad (1)$$

$$m(x) = E(f(x)) \quad (2)$$

$$K_f(x, x') = E[(f(x) - m(x))(f(x') - m(x')))] \quad (3)$$

where GP serves as a function prior for the process parameter variables. In this study, we employ the following covariance function to represent the covariance between random variable pairs:

$$K_f(x, x') = \sigma_f^2 \exp(-(x - x')^T M (x - x') / 2) \quad (4)$$

where σ_f^2 is the signal variance and $M = \text{diag}(l^{-2})$ is a diagonal matrix with the length-scale vector l . Note that $f(x)$ and $f(x')$ should be similar if x and x' are sufficiently close in the temporal dimension. Therefore, the length scale l defines the separation between different dimensions of input variables. For a missing data y_* at the input x_* , the training outputs $Y = f(X)$ from $X = (x_1, \dots, x_n)$, and the predicted $y_* = f(x_*)$ have a joint prior distribution with zero mean:

$$\begin{bmatrix} Y \\ y_* \end{bmatrix} \sim \mathcal{N} \left(0, \begin{bmatrix} K_f(X, X) + \sigma_n^2 I & K_f(X, x_*) \\ K_f(x_*, X) & K_f(x_*, x_*) \end{bmatrix} \right) \quad (5)$$

Additionally, the hyper-parameters $\theta = \{M, \sigma_f, \sigma_n\}$ need to be optimally chosen in order to yield the best GP model for predicting the imputed values. These hyper-parameters can be learned by maximizing the log marginal likelihood,

$$\theta_{\text{optimal}} = \arg \max_{\theta} \{\log p(Y|X, \theta)\} \quad (6)$$

$$\begin{aligned} \log p(Y|X, \theta) &= -\frac{1}{2} \log |K_f + \sigma_n^2 I| \\ &\quad -\frac{1}{2} Y^T (K_f + \sigma_n^2 I)^{-1} Y - \frac{n}{2} \log(2\pi) \end{aligned} \quad (7)$$

2.3. Multiobjective optimization

2.3.1. Pareto optimal solution set

The aim of this research was to simultaneously maximize the strength and modulus of carbon fiber-reinforced EP composite materials. To address these distinct objectives simultaneously, multiobjective optimization was conducted. Given that improving one objective often influences another, a singular solution to this optimization problem does not exist. Instead, a range of solutions exist along the Pareto frontier, known as nondominated solutions [36,37]. These Pareto frontier solutions allow designers to identify the most favorable design solutions that effectively balance multiple objectives.

This study employed a variation of genetic algorithms known as NSGA II for multiobjective optimization [35,38,39]. NSGA II uses a fast-elitist selection strategy to search for the Pareto optimal solution set, selecting solutions from different fronts on the basis of their ranking and crowding distance. During the nondominated sorting process in NSGA II, the algorithm identifies and ranks nondominated solutions in the population (Fig. 4). This classification is based on their dominance levels and crowding distances.

The resulting set of nondominated solutions represents the Pareto frontier, which denotes the optimal trade-off curve, signifying that the enhancement of one objective would require compromises in another.

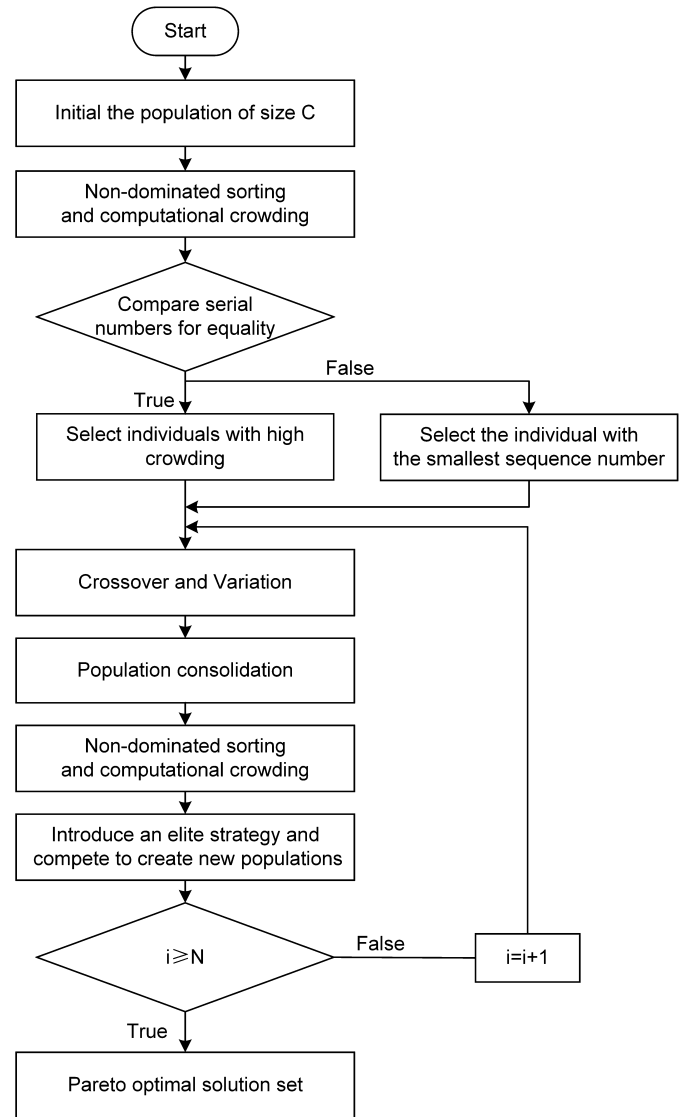


Fig. 4. Flowchart for NSGA II algorithm.

NSGA II commences with a randomly selected population and introduces elitism after the initial generation. This process involves comparing the population with the most favorable nondominated solutions identified previously. Subsequently, the algorithm employs binary selection, based on nondominance and crowding distance, to generate a new population. This stage is followed by the processes of crossover and mutation. Crowding distance plays a crucial role in ranking solutions, as it measures the density of solutions unrounding a particular point. It does so by measuring the extent of unoccupied objective space surrounding a solution in relation to other solutions in the population.

2.3.2. Combinatorial weighting on the basis of game theory

At present, the field of materials lacks an authoritative weighting evaluation metric for measuring the performance of composite materials. Consequently, performance requirements are treated as subjective weighting criteria and are objectively assessed using the combination ordered weighted averaging (C-OWA) operator [40]. To mitigate potential biases stemming from relying solely on subjective and objective weightings for determining weights and to more accurately reflect real-world scenarios, the concept of game theory was introduced. We established a game set model aimed at minimizing the deviation in composite weights [41,42].

Step 1: Determine the weight of each index through two different methods: subjective weight (w_1) and objective weight (w_2).

Step 2: Construct the combined weight w^* , which is any linear combination of the two weight vectors:

$$w^* = \alpha_1 w_1^T + \alpha_2 w_2^T \quad (8)$$

where α_1 and α_2 represent the linear combination coefficients of subjective and objective weights, respectively.

Step 3: Implement a game-theoretic approach between the weight vectors obtained from the two methods, optimizing the parameters α_i of the linear combination coefficient. The goal is to minimize the dispersion result between the combined weight w^* and each weight w_k , thereby deriving the most satisfactory combined weight value:

$$\min \|w^* - w_k^T\| \quad (k = 1, 2) \quad (9)$$

where $\|\cdot\|$ represents the matrix 2-Norms. The constraint equation for optimizing the first derivative of Equation (9) is as follows:

$$\sum_{f=1}^2 \alpha_f w_k w_f^T = w_k w_k^T, k = 1, 2 \quad (10)$$

Equation (10) can be reformulated into an equivalent linear equation system as follows:

$$\begin{bmatrix} w_1 w_1^T & w_1 w_2^T \\ w_2 w_1^T & w_2 w_2^T \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} = \begin{bmatrix} w_1 w_1^T \\ w_2 w_2^T \end{bmatrix} \quad (11)$$

Step 4: Identify the linear combination coefficient vector and normalize the result:

$$\alpha_k^* = |\alpha_k| / \sum_{k=1}^2 \alpha_k \quad (12)$$

At this stage, the combined weight of the evaluation index based on game theory is obtained as follows:

$$w^* = \alpha_1^* w_1^T + \alpha_2^* w_2^T \quad (13)$$

2.3.3. Decision-making based on combinatorial weighting and TOPSIS

In this study, the combination of the TOPSIS with NSGA II was employed to enhance the understanding of the solution space and to enable the selection of the optimal solution [43,44]. TOPSIS was used to consolidate multiple objective responses into a singular criterion, thereby enabling the evaluation of a finite set of solutions based on various criteria. By amalgamating the TOPSIS with NSGA II, an effective method emerges for choosing the best solution from the Pareto optimal set, a task which was challenging when relying solely on NSGA II. This method involved ranking the Pareto optimal solutions on the basis of multiple criteria, focusing on the selection of the solution that most closely aligned with the ideal solution. The ranking process included evaluating proximity to both the positive ideal solution and the negative ideal solution, thereby providing a relative measure of closeness to the ideal solution.

The steps of the constructed game theory combinatorial weighting-TOPSIS evaluation model were as follows:

Step 1: Construct the weighted and normalized decision matrix.

The weighted normalized decision matrix C_{ij} is obtained by multiplying the normalized matrix T_{ij} with the combination-weighted weight vector w^* :

$$C_{ij} = (c_{ij})_{mn} = \begin{pmatrix} w_1 t_{11} & w_2 t_{12} & \dots & w_n t_{1n} \\ w_1 t_{21} & w_2 t_{22} & \dots & w_n t_{2n} \\ \dots & \dots & \ddots & \vdots \\ w_1 t_{m1} & w_2 t_{m2} & \dots & w_n t_{mn} \end{pmatrix} \quad (14)$$

where T_{ij} is the normalization of the Pareto optimal solution.

Step 2: Determine positive and negative ideal solutions.

The evaluated positive ideal solution C_j^+ and negative ideal solution C_j^- can be expressed as follows:

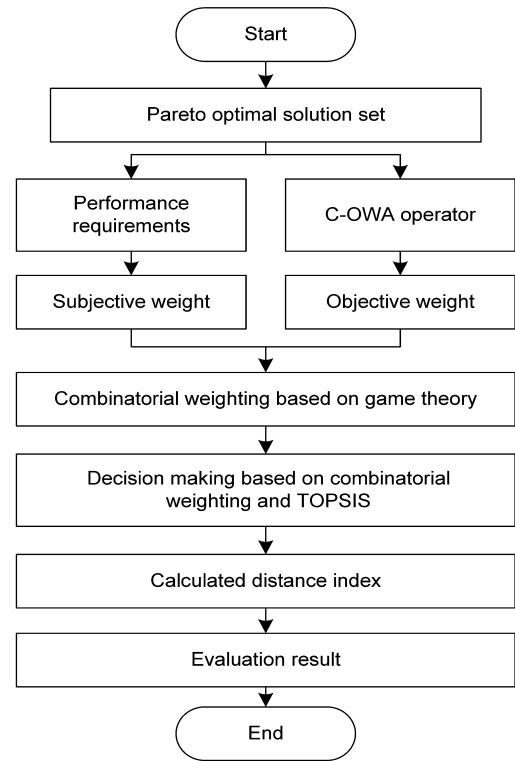


Fig. 5. Comprehensive credit evaluation process based on combinatorial weighting and TOPSIS method.

$$C_j^+ = (C_1^+, C_2^+, \dots, C_n^+) = \max_{1 \leq i \leq m} c_{ij} \quad (15)$$

$$C_j^- = (C_1^-, C_2^-, \dots, C_n^-) = \min_{1 \leq i \leq m} c_{ij} \quad (16)$$

Step 3: The distance between the evaluated performance and the positive and negative ideal solutions is calculated using Euclidean distance:

$$D_i^+ = \sqrt{\sum_{j=1}^m (c_{ij} - C_j^+)^2} \quad (17)$$

$$D_i^- = \sqrt{\sum_{j=1}^m (c_{ij} - C_j^-)^2} \quad (18)$$

Step 4: Calculate the proximity index.

$$R_i = D_i^- / (D_i^+ + D_i^-) \quad (19)$$

The proximity index, which ranges from 0 to 1, quantitatively reflects how close the evaluated metric is to the positive and negative ideal solutions. A proximity index approaching 0 suggests that the performance metric is leaning toward the negative ideal solution, indicating inferior evaluation results. Conversely, a proximity index approaching 1 signifies that the performance metric is moving toward the positive ideal solution, indicating superior evaluation results. Fig. 5 illustrates the comprehensive evaluation process of the game theory combinatorial weighting and TOPSIS method.

3. Results and discussion

3.1. Materials

3.1.1. Fourier transform infrared spectrum

The Fourier transform infrared spectrum of the aminated carbon fiber was shown in Fig. 6. The spectra of CF show characteristic peaks of the -OH group at 3437 cm⁻¹ and the -C=O group in -COOH at 1637 cm⁻¹. It can be inferred that CF has been successfully oxidized. Compared with the infrared spectrum of CF, the infrared spectrum of CF/KH-540

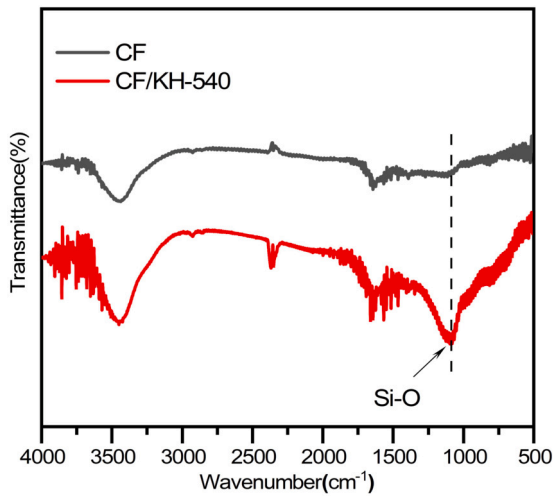


Fig. 6. Fourier transform infrared results for carbon fiber.

has a new absorption peak at 1086 cm⁻¹. The characteristic absorption peak of -Si-O indicates that KH-540 has formed a covalent bond with the surface of carbon fibers. Infrared spectroscopy confirmed that the amino groups were successfully loaded on the carbon fibers.

3.1.2. Mechanical properties

The effect of surface treatment, including oxidation treatment and the addition of silane coupling agent, on the tensile properties of CFRE is depicted in Fig. 7(a). The tensile strength of CFRE-P was 54.05 MPa. The tensile strength of CFRE-Si was slightly lower (49.06 MPa). The tensile strength of CFRE-O-Si was 55.9 MPa. These results suggest that oxidation treatment improves the effectiveness of the silane coupling agent. Notably, the addition of the silane coupling agent without oxidation treatment decreased tensile strength. This variation in the reinforcing effect of the silane coupling agent, dependent on the presence of oxidation treatment, is an indication of the complexity introduced by multiple factors in the curing process.

The effects of carbon fiber on the tensile and compressive strengths of CFRE are illustrated in Fig. 7(b). Carbon fibers increased the compressive strength of the EP by 17.8% to 84.8 MPa. However, carbon fiber decreased the tensile strength of the EP by 12.5% to 55.9 MPa. These results indicate that a single variable can have different effects on different performance characteristics.

Accordingly, although experiments can offer preliminary insights into correlations between parameters and performance, they are insufficient for gaining comprehensive insights. Multivariate collaborative optimization experiments are essential for more nuanced analysis.

3.2. Experimental design and multifactor test

3.2.1. Multifactor parameters

Four inputs were selected based on relevant literature and preliminary experiments, including carbon fiber content (CFC) which ranging from 0 to 8%, secondary curing heating temperature (ST) which ranging from 60 to 130 °C, secondary curing heating time (St) which ranging from 0 to 5 hours, and silane coupling agent solution concentration (Sc) which ranging from 0 to 10%. The input ranges were presented in Table 1. The last column of Table 1 represents the number of intervals for each parameter in the Latin hypercube sampling method used in our study.

3.2.2. Multifactor test

For the multifactor test, 23 experimental points were selected on the basis of the Latin hypercube point selection principle, considering four input and four output variables. The mechanical performance test

Table 1

Latin hypercube sampling parameters.

Parameters	Minimum	Maximum	Layered
CFC (%)	0	8	9
ST (°C)	60	130	15
St (h)	0	5	21
Sc (%)	0	10	6

results for these 23 experimental points are illustrated in Table 2. Ts, Tm, Cs, and Cm represent tensile strength, tensile modulus, compressive strength, and compressive modulus, respectively.

Scatter graph of process parameters (i.e., CFC, ST, St and Sc) against mechanical properties (i.e., Cm, Cs, Tm and Ts) is presented in Fig. 8.

A notable observation is the behavior of these properties in relation to the variation in input parameters. As CFC increases, tensile strength initially decreases, then increases, and then finally decreases again; tensile modulus tends to remain stable, as illustrated in the first column of the plot. Compressive strength and compressive modulus exhibit an initial increase, followed by a decrease, and then an increase. Conversely, an increase in Sc leads to an initial decrease and then an increase in compressive strength and compressive modulus, whereas tensile strength and tensile modulus do not exhibit a clear trend, as illustrated in the fourth column. The experimental points are evenly distributed across the plot, indicating no significant trends for the variables represented in the second and third columns.

The distribution of mechanical performance points suggests a nonlinear variation. This necessitates a deeper analysis of the relationship between input and output parameters.

To investigate the relationships between input and output variables, Spearman's rank correlation analysis was performed. The results of this analysis are shown in Fig. 9. Tensile strength was inversely correlated with ST and CFC. Tensile modulus was inversely correlated with St and weakly correlated with the other input variables. Compressive strength was influenced by St. Compressive modulus exhibited a medium positive correlation with Sc and St.

A nonlinear correlation between the input and output variables is evident (Fig. 8). Each input and output variable has a different correlation. This suggests that different enhancement mechanisms for different performance aspects could result in conflicts during optimization. The improvement of tensile strength is mainly attributed to the formation of long chains through high molecular polymerization. The improvement in tensile modulus is dependent on the pull-out strength of carbon fiber and the interfacial bonding strength between carbon fibers and epoxy [45]. Compressive strength is primarily enhanced by promoting the increase in crosslinking density through the formation of more covalent crosslinking bonds at high temperatures. The enhancement of compressive modulus is mainly through the increase in crosslinking density and interface strength.

Therefore, when multiple response targets are involved in material reinforcement, conflicts among the strengths of each target are inevitable. To navigate these conflicts, a multiobjective optimization scheme is essential to balance the advantages of each objective effectively.

3.3. Multiobjective optimization

3.3.1. Evaluation of GP model

The construction of the Gaussian model, we refer to the multivariate Gaussian model by Sadoughi et al. [46], in which the kernel function of the selected Gaussian model is Squared Exponential. The hyperparameter optimization is carried out by applying fmincon function to minimize the negative log-likelihood, and the optimal values of the hyperparameters are respectively $\sigma_f = [1, 1, 1, 1]$, $\sigma_n = [1, 1, 1, 1]$, $l = [0.3288, 2.6631, 1.3572, 1.6681]$. In addition, the required parameters for the construction process of the Gaussian model were set as $Model.s = 0.6$,

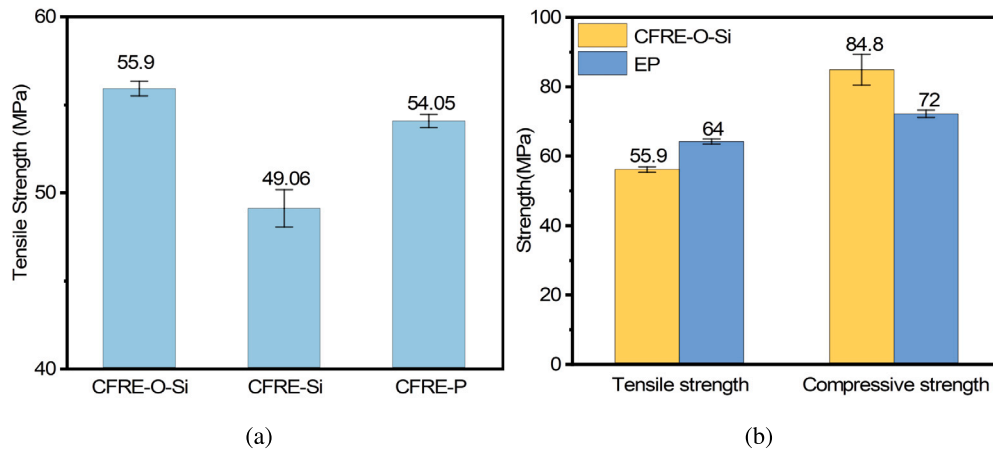


Fig. 7. (a) Tensile strengths of samples with different surface treatment. (b) Tensile and Compressive strengths of samples with different carbon fiber concentration.

Table 2

Experimental results from Latin hypercube sampling.

No.	Input parameters				Output parameters			
	CFc (%)	ST (°C)	St (h)	Sc (%)	Ts (MPa)	Tm (MPa)	Cs (MPa)	Cm (MPa)
1	1	70	1.25	10	53.93	1978	74.77	2382
2	3	95	3.25	10	41.25	1571	74.1	2470
3	2	130	3.75	10	61.3	1593	77.47	2281
4	2	85	2.25	2	56.43	2692	77.17	2166
5	1	115	2.5	0	56.36	1577	74.43	2414
6	7	110	4	10	45.85	2112	80.47	2518
7	8	70	2.5	4	46.55	1782	66.3	2224
8	2	110	4.5	6	51.8	1917	79.57	2292
9	5	65	5	4	52.65	2167	67.93	2295
10	4	75	0.75	0	55.7	1478	70.7	1513
11	3	100	0.5	8	38.6	1456	78.37	2588
12	6	95	0	4	61.9	1836	70.33	2123
13	8	130	3.5	0	41.55	1594	81	1915
14	3	125	4.25	2	35.07	1653	79.7	2505
15	5	125	1	4	50.7	1682	75.6	2056
16	7	120	2	8	49.53	1710	70.03	2503
17	5	60	2.75	8	56.4	1949	72.57	2479
18	6	80	3	2	52.33	1676	77.3	2354
19	1	90	3.75	4	58.9	1583	76.1	2150
20	6	100	1.5	0	47.33	1823	73.53	2281
21	4	65	5	8	58.83	1755	72.47	2599
22	8	105	1.25	6	48.27	1877	77.9	2410
23	7	80	4.75	10	53.45	1896	77.37	2504

$Model.degree = 0$, $Model.optim = fmincon$; Where, $Model.s = 0.6$ was determined according to the Spearman covariance value, with the default values adopted by $Model.degree$ and $Model.optim$.

The accuracy of the GP model was evaluated to confirm the validity of its predictive capability. Three metrics were employed for this evaluation: mean absolute error (MAE), mean squared error (MSE), and root mean squared error (RMSE). MAE signifies the mean of the absolute differences between predicted and actual values. MSE denotes the average of the squared deviations between the predicted and actual values. RMSE quantifies the deviation between the predicted and actual values. The specific evaluation criteria are presented in Table 3.

The accuracy of the model was affected by hyperparameters. The accuracy of the model was evaluated by monitoring the iteration of values during optimizing hyperparameters. A tendency of these values to stabilize suggested the completion of hyperparameter optimization and indicated that the model had achieved relatively optimal accuracy. After approximately 15 iterations, the fitness curve stabilized, with significant deviation from the initial value, indicating that a relatively optimal set of hyperparameters had been discovered at that point. Stabilization occurred at a value of approximately 7 after approximately 170 iterations, representing a decrease from the initial value of 13 (Fig. 10),

Table 3

Evaluation of Gaussian process model.

Index	Mean value	MAE	MSE	RMSE
Ts	51.41	0.1704	0.0474	0.2178
Tm	1823	0.1845	0.0521	0.2283
Cs	74.86	0.1362	0.0298	0.1727
Cm	2321	0.1790	0.447	0.2113

indicating that the optimal point satisfying maximum likelihood had been found.

3.3.2. Pareto optimal solution set

The GP model was used as the optimization function. NSGA II was used for optimization, with the Pareto optimal solution set serving as the optimal solution.

Due to the stochastic nature of NSGA-II, we conducted ten repeated runs of the algorithm with fixed parameters, and the minimum crowding distance was used to evaluate the convergence. All of ten runs convergence within 200 generations. Thus, we set the generation number as 200 in this work. Results of Pareto set are presented in Table 4.

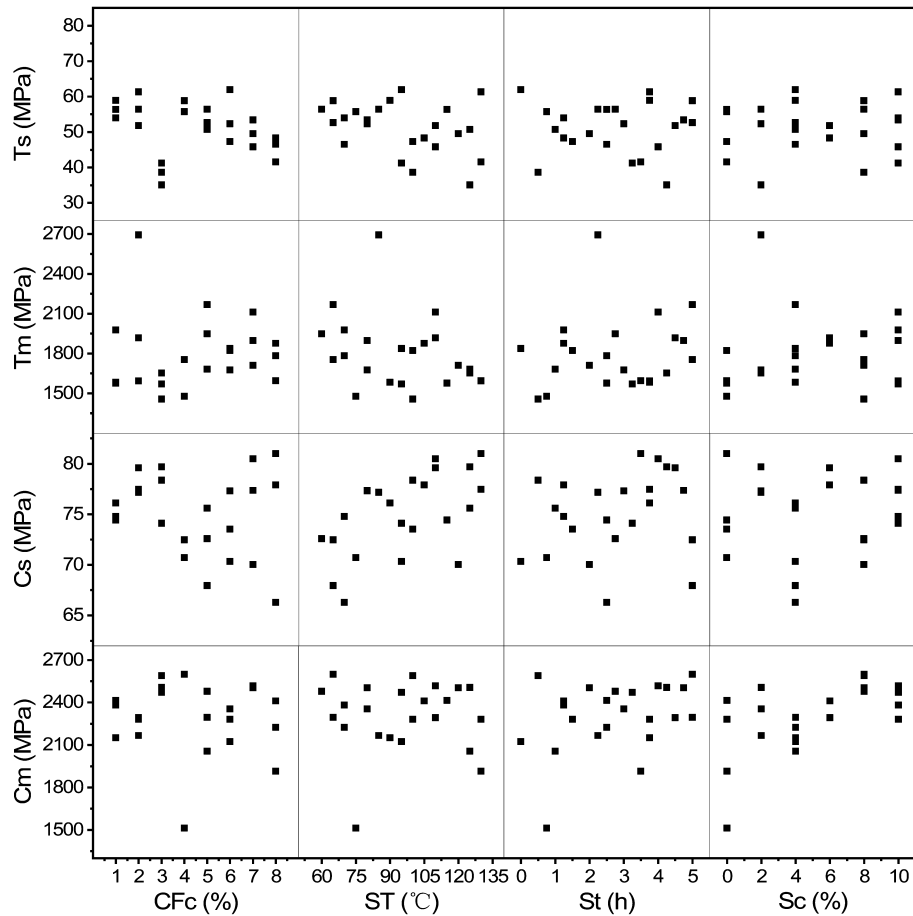


Fig. 8. Scatter graph of process parameters (i.e., CFc, ST, St and Sc) against mechanical properties (i.e., Cm, Cs, Tm and Ts).

Table 4
Pareto optimal solution set based on NSGA II.

No.	Input parameters				Output parameters			
	CFc (%)	ST (°C)	St (h)	Sc (%)	Ts (MPa)	Tm (MPa)	Cs (MPa)	Cm (MPa)
1	5	110	0.5	8	58.77	1546	77.01	2793
2	1	70	4.5	2	67.94	1920	79.11	2075
3	1	70	4.5	4	66.44	1831	70.9	2175
4	6	110	4.5	2	39.13	2447	83.24	2644
5	2	110	2	6	34.39	2684	81.41	2655
6	5	110	0.5	8	58.77	1546	77.01	2793
7	1	70	4.5	2	67.94	1920	79.11	2075
8	1	70	0.5	6	62.02	1681	79.25	1924
9	6	90	4.5	2	44.36	2467	82.5	2766
10	2	70	2	6	64.14	3012	78.68	2404
11	3	70	2	6	59.98	2826	77.37	2445
12	3	100	1.5	8	39.98	2101	75.66	2770
13	5	100	2	8	54.15	2011	73.92	2747
14	5	110	4.5	2	39.75	2326	82.81	2691
15	1	70	1.5	6	64.11	2891	80.15	2242
16	5	110	2	6	52.11	2240	79.15	2606
17	1	70	4.5	8	66.31	1684	72.9	2503
18	5	70	1.5	8	61.57	2126	77.47	2580
19	2	70	4.5	8	64.19	1727	72.78	2559
20	1	110	4.5	2	49.76	1892	81.24	2705
21	6	80	4.5	2	47.56	2435	81.47	2685
22	1	70	0.5	8	62.03	1615	81.76	2215
23	1	80	4.5	2	63.04	1938	80.15	2329
24	4	70	2	8	59.06	2353	76.39	2626
25	5	110	0.5	6	57.69	1724	80.73	2533
26	5	110	1	8	56.66	1597	74.92	2763
27	1	70	4.5	4	66.44	1831	70.9	2175
28	6	90	2	8	55.64	1952	74.21	2703
29	2	70	4.5	2	65.14	1998	79.35	2175
30	3	90	2	6	43.25	2726	79.83	2667

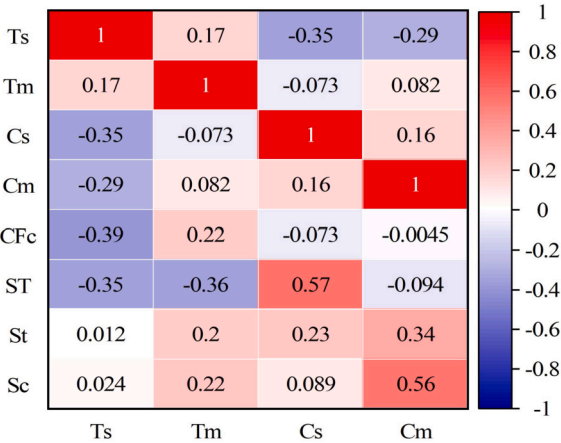


Fig. 9. Spearman's rank correlation matrix.

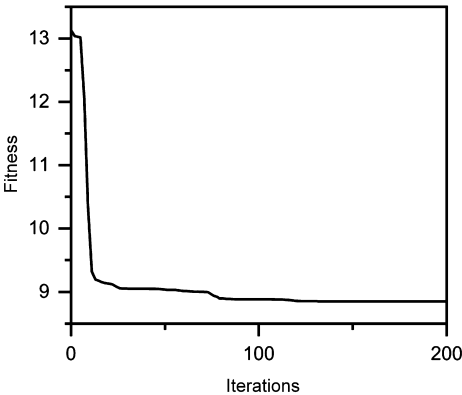


Fig. 10. Iterative convergence graph.

Table 5
Subjectivity weights based on linear weighting method and Objective weight based on C-OWA operator.

Weighting category	Output parameter priority	Weights ratio (Ts:Tm:Cs:Cm)
OCS	Cs > Ts > Cm > Tm	0.3:0.1:0.4:0.2
OCT	Cs = Ts > Cm > Tm	0.36:0.1:0.36:0.18
OEP	Ts = Cs = Tm = Cm	0.25:0.25:0.25:0.25
OSM	Cs = Ts > Tm = Cm	0.375:0.125:0.375:0.12
Objective		0.1413:0.1817:0.3421:0.3349

3.3.3. Combinatorial weighting on the basis of game theory

Subjective weights were assigned to the four output parameters (tensile strength, tensile modulus, compressive strength, and compressive modulus) on the basis of their relevance in different usage scenarios. In these weightings, compressive performance consistently received higher priority than tensile strength, and overall strength was prioritized over modulus. Consequently, four weight categories were established: optimized compressive strength (OCS), optimized tensile and compressive strength (OCT), optimized overall performance (OEP), and optimized strength with equal modulus (OSM). Objective weights for these four indicators (tensile strength, tensile modulus, compressive strength, and compressive modulus) were calculated using the C-OWA operator. These weight allocations are detailed in Table 5.

Subjective weighting aims to align indicator weight with actual importance by assigning weight order values to evaluation criteria on the basis of practical applications. However, this method can be heavily subjective and arbitrary. Conversely, objective weighting, derived from monitoring data, offers strong objectivity but may lack human insight and produce results that are not entirely consistent with real-world sit-

Table 6
Combinatorial weighting based on game theory.

Weighting category	Game Theory Weight		Composite Weight			
	subjectivity	objective	Ts	Tm	Cs	Cm
OCSGT	0.652924	0.347076	0.2449	0.1284	0.3799	0.2468
OCTGT	0.60732	0.39268	0.2741	0.1321	0.353	0.2408
OEPGT	0	1	0.1413	0.1817	0.3421	0.3349
OSMGT	0.634982	0.365018	0.2897	0.1457	0.363	0.2016

uations. Both subjective and objective weighting methods have their inherent limitations. Therefore, amalgamating the strengths of both methods through a careful analysis of specific material design processes is crucial. This approach ensures that subjective judgments and the objective importance of evaluation criteria are both considered.

The combinatorial weighting method, rooted in game theory, effectively integrates the characteristics of subjective and objective weighting. It strikes a balance between the subjective and objective weight vectors determined by various methods, using optimization theory to minimize the deviation between combinatorial and individual weights. This approach ultimately yields an optimal set of combinatorial weights, demonstrating high practicality for material design processes. In this paper, we employ this method to assign weights to the evaluation criteria.

On the basis of the aforementioned methodology, mixed weights were determined by using a game theory approach (Table 6). Four mixed weight categories, namely OCSGT, OCTGT, OEPGT, and OSMGT, were established using this game theory approach.

3.3.4. Multiobjective optimization result

The TOPSIS decision-making method was applied to rank the Pareto optimal solution set, yielding the top three solutions for various mixed weights (Table 7). The last column represents the degree of optimality based on the TOPSIS method for evaluating each solution in the Pareto frontier. The different mixed weights led to distinct rankings in the Pareto optimal solution sets. Generally, the solution with the highest optimal value is considered to correspond to the optimal solution under the respective weight category.

Interestingly, the optimal solutions associated with the weights OCSGT and OSMGT converged at the same point. Furthermore, with the exception of the weight OEPGT, the top three optimal solutions remained identical across the remaining weights. This uniformity in results is attributed to the coordinated and balanced application of subjective and objective weights through the mixed weight method rooted in game theory. This approach mitigated the inherent biases stemming from subjective weights while preserving their distinctive tendencies. Conversely, due to the uniform distribution of subjective weight in OEP, the weight OEPGT aligned with the objective weight following a reorganization using game theory. Consequently, compared with other optimal solutions, those based on the OEPGT weight exhibited significant differences, which significantly influenced the distribution characteristics of performance strength.

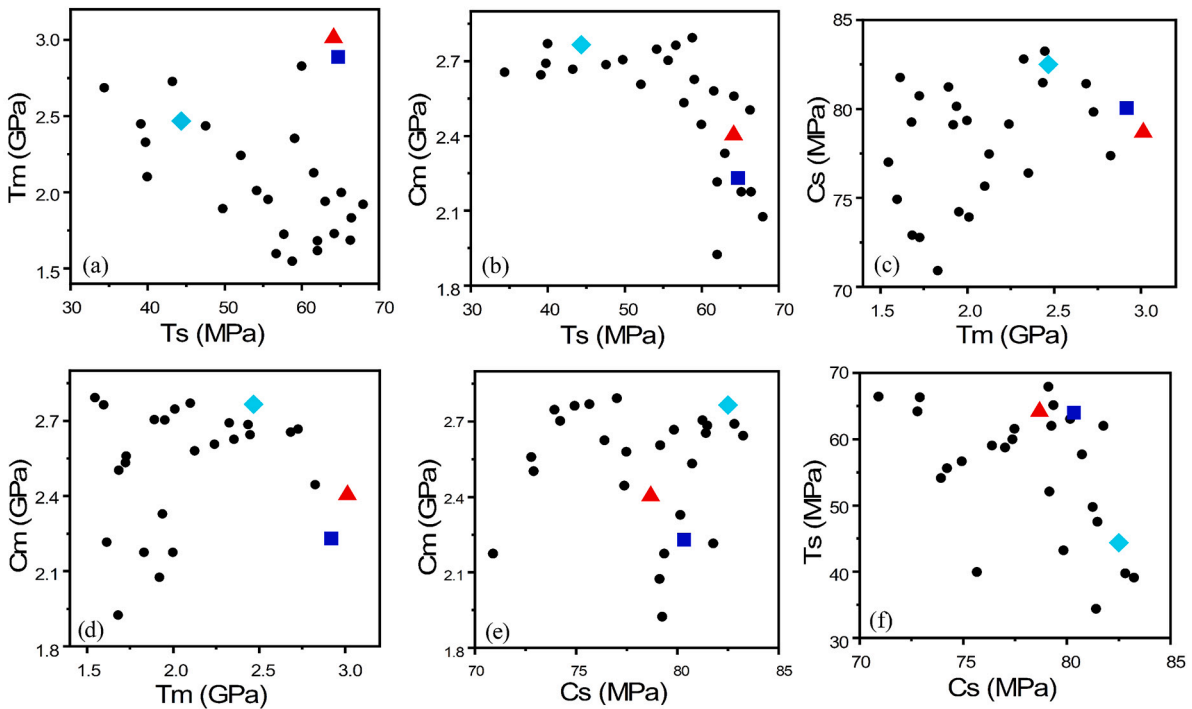
The application of mixed weights grounded in game theory effectively amalgamated the features of subjective and objective evaluation methods, averting scenarios in which objective weights deviated from practical applications. Furthermore, the adjustment of subjective weight offered the flexibility to tailor the optimal target performance distribution according to the diverse performance requirements of various application fields.

The optimal solution was not unique but rather represented by a Pareto frontier composed of 30 optimal solutions (Fig. 11). Among these solutions, the cyan rhombus represented the optimal solution obtained based on the weight OEPGT, the blue square represented the optimal solution based on the weight OCTGT (OSMGT), and the red triangle represented the optimal solution based on the weight OCTGT.

Table 7

Pareto optimal solutions based on different subjective weights set the first 3 solutions in order.

Weighting category	Order number	Input				Output				Optimality
		Cfc (%)	ST (°C)	St (h)	Sc (%)	Ts (MPa)	Tm (MPa)	Cs (MPa)	Cm (MPa)	
OCSGT	1	1	70	1.5	6	64.11	2891	80.15	2242	0.7160
	2	2	70	2	6	64.14	3012	78.68	2404	0.7116
	3	1	80	4.5	2	63.04	1938	80.15	2329	0.6759
OCTGT	1	2	70	2	6	64.14	3012	78.68	2404	0.7281
	2	1	70	1.5	6	64.11	2891	80.15	2242	0.7274
	3	1	80	4.5	2	63.04	1938	80.15	2329	0.6820
OEPGT	1	6	90	4.5	2	44.36	2467	82.50	2766	0.7390
	2	6	80	4.5	2	47.56	2435	81.47	2658	0.7192
	3	6	110	4.5	2	39.13	2447	83.24	2644	0.6942
OSMGT	1	1	70	1.5	6	64.11	2891	80.15	2242	0.7557
	2	2	70	2	6	64.14	3012	78.68	2404	0.7434
	3	1	80	4.5	2	63.04	1938	80.15	2329	0.6908

**Fig. 11.** Optimal solutions of graphs with two-dimensional presentation based on different weights, s , include OCSGT and OSMGT (■), OCTGT (▲), and OEPGT (◆).

Analyzing the optimal solutions derived from various weights revealed that the solution associated with the OEPGT weight stood out, exhibiting higher compressive strength and modulus (Fig. 11 (e)). The optimal solutions, based on the weights OCSGT and OSMGT, were identical and situated at the same position. Furthermore, the optimal solutions, based on the weights OCTGT and OCSGT (OSMGT), were centrally distributed in terms of location and each presented slightly varied performance characteristics. Specifically, the solution derived from OCTGT demonstrated a higher compressive strength (Fig. 11(f)), as well as lower compressive modulus and tensile modulus (Fig. 11(d)) when compared with the solution based on OCSGT (OSMGT). The weights OCTGT and OCTGT (OSMGT) exhibited similar expressions in tensile strength (Fig. 11(a)). Such variations in the distribution of optimal solutions reflected differences in performance attributes, with the influence of different weights influencing the performance outcomes. While optimal solutions depicted in the two-dimensional graph were not all non-dominated, those selected based on their weights tended towards higher performance, albeit with some underperformance in specific scenarios. For example, the optimal solution derived from the OCTGT and OCSGT (OSMGT) weights was situated at the center of the Cs-Cm in-

teraction (Fig. 11(e)), whereas the solution optimized with the OEPGT weight was located amidst the points representing the Ts-Tm interaction (Fig. 11(a)). Such a spatial arrangement of these solutions suggested that they were suboptimal within the entire set of optimal solutions, deviating from our intended standards.

The optimal solution was located on the Pareto front (Fig. 12). This front was visualized in the diagram using four dimensions: abscissa, ordinate, the size of the graph elements, and color. The graphical representation used the horizontal axis for tensile strength and the vertical axis for compressive modulus. The color represented compressive strength, and the size of the shapes indicated tensile modulus. The 30 optimal solutions were all non-dominated solutions, collectively forming the Pareto frontier. Across different dimensions, optimal solutions based on weights OCTGT and OCSGT (OSMGT) exhibited similarity, while those based on weight OEPGT demonstrated significant differences. It could be inferred that this was due to varying weight selections. Weight OEPGT reflected objective weights, whereas weights OCTGT and OCSGT (OSMGT) represented a mixture of objective and subjective weights. Thus, optimal solutions based on weights OCTGT and OCSGT (OSMGT) showed similarities in performance but still ex-

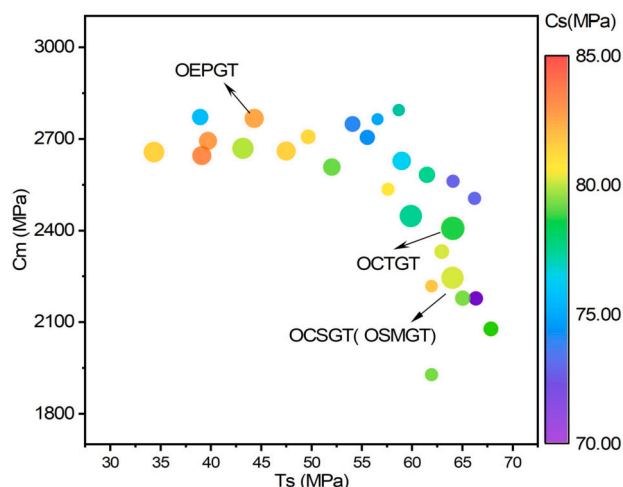


Fig. 12. Optimal solution based on different weights.

hibited some variance. Therefore, the choice of an optimal solution was fundamentally determined by the specific performance requirements of the sample.

In general, the distribution of the optimal solution is closely tied to the assigned weights, demonstrating a direct correlation between the weight and the resultant performance characteristics. Consequently, the distribution of the optimal solution can be effectively altered by adjusting the mixed weights.

Evaluate Pareto solutions based on different weights and calculate optimal value (Table 7). The solution with the maximum optimal value was considered the optimal solution. The optimal solutions at the Pareto front (OCSGT, OCTGT, OEPGT, OSMGT) were experimentally verified. The optimization results conducted across different mixed weights and compared with experimental data are shown in Fig. 13. These findings indicate a strong correlation between the optimization predictions and the experimental results. Specifically, in Fig. 13(a) and (d), the average differences for tensile strength, tensile modulus, compressive strength, and compressive modulus between optimized and experimental results were approximately 3.4%, 4.9%, 2.5%, and 3.5%, respectively. Similarly, Fig. 13(b) displayed differences ranging from 0.6% to 5.3%. Fig. 13(c) illustrates differences in optimization and experimental results of approximately 2.4%, 12.9%, 4.6%, and 8% for tensile strength, tensile modulus, compressive strength, and compressive modulus, respectively.

The experimental results underscore the value of incorporating machine learning methods into material design. Each optimization scheme demonstrated slight fluctuations with changes in weight allocation, yet maintained high performance based on subjective weight distribution (Fig. 13). A high compressive strength and compressive modulus are evident in Fig. 13(c), albeit with a reduced tensile strength performance, attributable to the even distribution of subjective weights. Additionally, experimental validation of the optimal solutions based on OCSGT and OCTGT shares similarities (Fig. 13a and 13b). However, when compared with the optimized solutions based on OCSGT, those based on OCTGT were observed to have higher compressive strength and compressive modulus, but lower tensile strength and tensile modulus. Owing to the inherent accuracy limitations of the prediction model, the optimized results are approximations and may not perfectly represent the optimal solution. This limitation is evident in Fig. 13(a) and 13(d), where the experimental verification indicates identical outcomes for solutions based on OCSGT and OSMGT.

When undertaking multiobjective optimization, inherent conflicts between objectives often impede achieving a solution that excels across all performance metrics. Therefore, by assigning importance to each goal based on performance requirements and appropriately balancing

these objectives, an optimal solution that satisfies the various criteria can be achieved.

4. Conclusion

This study introduced a multiobjective optimization method for CFRE based on machine learning techniques. This method incorporates machine learning into material design and optimization, setting it apart from other studies by requiring less initial data and effectively combining algorithm optimization with experimental design. The utilization of LHS and GP modeling for small-sample prediction significantly reduced the requirement for repetitive experiments in material design and optimization. The efficient multiobjective optimization of CFRE was achieved by employing NSGA II to obtain the Pareto optimal solution set and further refining the model through game theory and the TOPSIS method.

The main conclusions are as follows:

- (1) An efficient decision-making process for multiobjective optimization was established, using a hybrid weighting approach combining game theory and the TOPSIS method to efficiently rank the Pareto optimal solution set and identify the optimal solution.
- (2) A multiobjective optimization process, incorporating the combined strengths of GP modeling and NSGA II, was successfully established for optimizing CFRE properties. This optimization strategy combines GP modeling and NSGA II to extend design methods for small samples and materials with multiple attributes.
- (3) The confluence of experimental design principles and algorithmic frameworks culminates in a potent strategy for material performance prognostication, particularly small samples and materials with multiple attributes. The experimental verification and the four optimization results are consistent, indicating that the optimization strategy is effective. The average differences between the optimal solutions based on OCSGT (OSMGT) and the experimental values for tensile strength, tensile modulus, compressive strength, and compressive modulus were approximately 3.4%, 4.9%, 2.5%, and 3.5%, respectively. When OCTGT was applied, the average differences ranged from 0.6% to 5.3%. In the case of OEPGT, the variances were approximately 2.4%, 12.9%, 4.6%, and 8% for tensile strength, tensile modulus, compressive strength, and compressive modulus, respectively.

CRediT authorship contribution statement

Si Zhang: Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization. **Chaoshuai He:** Writing – original draft, Validation, Methodology, Investigation, Conceptualization. **Zifan Wang:** Methodology, Investigation, Conceptualization. **Chuanbo An:** Validation, Methodology, Investigation. **Yun Chen:** Writing – review & editing, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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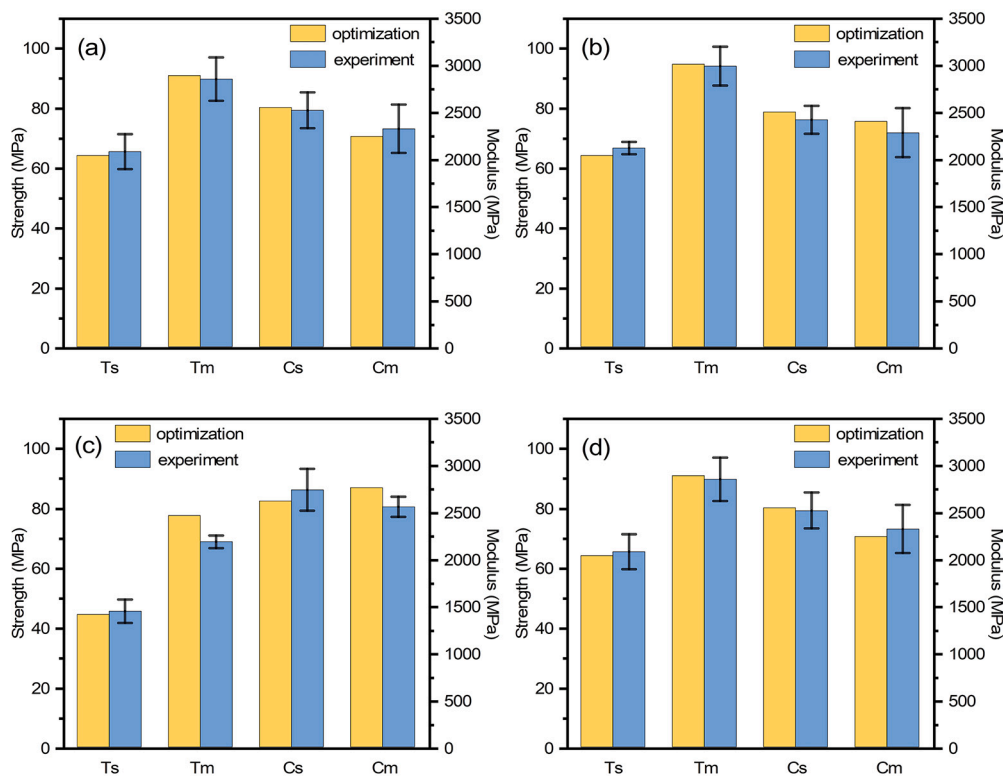


Fig. 13. Verification of optimization results based on (a) OCSGT, (b) OCTGT, (c) OEPGT, (d) OSMGT.

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